

THE EMERGING MARKET FOR FRUGAL INNOVATION: FAD, FASHION, OR FIT?

Working Paper

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ABSTRACT

There is growing interest among both private and public sectors to serve the underserved in emerging or developing countries leading to, what we call the market for frugal innovation. This paper is divided into two parts. First, we discuss the rhetoric surrounding frugal innovation and attempt to understand the discourse surrounding it as fad, fashion, or fit. Second, we seek to map the emergence of this market by suggesting drivers and the making of a social movement involving different actors pursuing both contentious and complementary approaches to achieve the same outcome, i.e. one of creating value for underserved populations. An understanding of how the rhetoric and market for frugal innovation has emerged will be useful in opening a research agenda. Consequently we cover opportunities and challenges to growth of the market as well as draw implications to academic theory and practice.

Keywords:

Frugal innovation, social movement, emerging market, constraints, fad, fashion.

Word Count: 9,450

INTRODUCTION

Popular discourse around ‘frugal innovation practices’ proposes new forms of innovation strategies and entrepreneurship meant to alleviate poverty in developing countries by improving markets and consumer experiences. We see a growing call for business, non-profits, and government to work together to tackle global concerns for poverty, inequality, and climate change. Combined with changes in the global economic landscape marked by the growth of emerging markets and the financial turmoil in developed markets, this offers a new space with new products, new competitors and collaborators.

Firms, business elites, and scholars have called for a wave of ‘rethinking business’ (Ventresca and Nicholls, 2011). Pfizer talked about its new emerging markets strategy as “devoted to meeting the diverse medical needs, across all disease areas, of patients in Emerging Markets around the world in an innovative, socially responsible and commercially viable manner” (Pfizer, 2011: online). Bill Gates, the richest and world's largest philanthropist, called on graduates of Harvard's 2007 class to invent "a more creative capitalism" that stretches “the reach of market forces so that more people can make a profit, or at least make a living, serving people who are suffering from the worst inequities” (para 26). Clayton Christensen et al (2006) suggest that instead of falling in the trap of social investments ending up maintaining the status quo, funds should help create “catalytic innovations” aimed at resolving the needs of underserved segments. Robert Guest (2011), Global Business Editor for The Economist, argues in “Borderless economics: Chinese sea turtles, Indian fridges, and the new fruits of global capitalism” how Indian diasporas' frugal innovations like \$70 refrigerators and \$2,000 cars and associated managerial knowhow can reshape the global economy as these technologies find their way to the West.

The diverse products and services purported to represent ‘frugal’, ‘inclusive’, ‘catalytic’, ‘trickle-up’, ‘reverse’, ‘BOP’ etc. in both practitioner and academic literature are outlined in table 1.

The solutions are often characterized by constraints on inputs and outputs such as price, performance, accessibility, availability, usability, and sustainability.

Insert Table 1 here

The practice of engineers working with constraint is not new. But we argue the form of constraints has shifted in the current period: Away from pure efficiency or even technical precision, to a ‘mission-driven’ set of constraints. There have always been engineers tinkering with improved yet simpler ways to do things. Some corporations began to explore a purposeful use of this recurring habit, sending group of engineers to various settings in Africa and elsewhere with a mandate to ‘do something, learn something’. Debra Dunn in her HP days, for example, pioneered sending teams of 30+ HP engineers on such quests.

But unlike NASA, the ‘mission’ here is mundane: to create offerings for people who live and work at the least privileged interfaces of the modern world. And this is the ‘why’ of frugal innovation: A broadly shared view to provide consumer value/amenities with the feature of ‘extreme affordability.’ These may be due to philanthropic, market or efficiency seeking reasons. There are dozens of these initiatives, in every sphere and sector and only some go by the actual term ‘frugal innovation.’ So the ‘what’ is varied. The variety of terms suggests the variety of processes involved: ‘Innovation under constraint’, ‘reverse engineering’, ‘frugal engineering’, etc. But the ‘how’ is also varied. Analytically the ‘what’ is all over the map, the ‘how’ is a smaller set of activities, and the ‘why’ is an even narrower set of reasons (table 2). We sketch out some arguments to help make sense of the variety of practices and images that comprise this semantic web of talk and the underlying activities that give it heft. We need to understand the variety of meanings and usages in the term ‘frugal innovation.’ But to get started, we have to recognize that the activity of frugal innovation is not entirely new though it has and is evolving.

Insert Table 2 here

This paper attempts to advance our understanding of the evolving rhetoric in ‘rethinking business’ and one particularly focused on ‘doing more with less’ for both producers and consumers. The varied jargon may each constitute fad-like popularity but we argue that when considered in unison, there is evidence that the rhetoric will persist as an overall fashionable trend and likely evolve to realize better ‘fit’ for implementation. In an attempt to understand the market for frugal innovations we offer three main factors that have helped raise awareness of the (frugal) market’s development. Markets form by technology push (Geroski, 2003), demand inducement (Prahalad, 2005), political approach (Fligstein 1996) in setting boundaries (Santos & Eisenhardt, 2009), intermediaries (Mair, Marti and Ventresca, 2011), and social movements (Hargrave & Ven, 2006; Zald, Morrill, & Rao, 2005). In the last three decades there has been simultaneous growth in emerging markets, information and communications, and social movements. We offer these three main drivers as having helped raise awareness for the creation of the market for frugal innovations.

GROWING PUBLICATIONS ON THE ‘WHAT’

Many people claim they are practising frugal innovation or something close to it. As the field evolves we will see an even greater set of new experts exhibiting authoritative rhetorical talk and buzzwords. We observe that the discourse has transcended the interfaces between journals of development, engineering, and management. For instance, development economists discuss appropriate technology, engineers discuss lean engineering, financial specialists discuss investment with impact, and policy analysts discuss public-private collaboration for poverty or inequality reduction.

A bibliometric search of Google Scholar was made on all the concepts listed in column 1 of Table 2 since 1990 and the findings are presented in figures 1 through 3. Google Scholar was used for the search given its growing popularity in revealing scholarly literature comprising peer-reviewed papers, theses, books, preprints, abstracts as well as technical reports from practitioners and policy makers representing broad areas of research. Additionally, Google Scholar has the ability

to extract references in several languages. Concepts which resulted in less than twenty publications since 1990 are not shown in the plots as they were deemed not popularly significant enough.

Insert Figure 1 here

Insert Figure 2 here

We see the most popular concepts are reverse engineering, appropriate technology, and lean engineering-manufacturing-product development, mainly because they have been around since the 1960s and 1980s. So we categorize these as precursors to the recognition of more related mission driven constraints associated with ‘frugal’, ‘inclusive’, ‘catalytic’, ‘trickle-up’, ‘reverse’, and ‘BOP’ innovations.

“Reverse engineering” is a technique for revealing a product’s features, technologies, and re-manufacturing. Although it could be a method used by many emerging market firms to achieve ‘frugal innovation’, its use and practice extends far beyond that sole purpose. The lean concepts stem from the Japanese management and Total Quality Management literature. Though the ‘lean’ literature may have relevance to apply lean solutions to achieve ‘inclusive’ or ‘frugal’ innovation, but is not defined by or limited to it.

“Appropriate technology”, according to (Kaplinsky, 2010), arose in the 1960s from a recognition that most global technological progress was directed to meet the needs of the global rich, and was designed to operate in developed high-income environments. Epitomised in "Small is Beautiful" (Schumacher, 1974), the spread of “appropriate technologies” was an agenda largely pursued by the not-for-profit Appropriate Technology movement. With mounting concerns on aid and hand-outs, the opening up India and China to the market mechanism, and the "rapid rise of incomes of the very poor – the “second bottom billion” – innovation for the poor and innovation appropriate for production in low-wage and poor-infrastructure environments has increasingly become an arena for profitable production" (Kaplinsky, 2010). The movement seems to have

readjusted given the lukewarm successes and failure of the appropriate technology concerns.

Though the methods have changed but the mission isn't very different.

In the same vein, Prahalad and Mashelkar (2010) argue that most innovation programs are built on the assumptions of affluence and abundance. However, they contend this is being challenged by contemporary notions of affordability and sustainability which are replacing premium pricing and abundance as innovation's drivers. As such they exhort firms to learn to do 'more with less' and for 'more people'. They contend that "a potent combination of constraints and ambitions has ignited a new genre of innovation" (p.3). We collate most of the highly touted 'frugal innovations' in table 1.

A challenge to the assumptions of affluence and abundance means that the worldwide risks of a professional-use driven innovation (top-down) strategy for the existing global multinational corporations have increased significantly. The huge research, development, prototype and global marketing costs, coupled with ever-increasing numbers of competing international players means that the amount of time during which a firm can enjoy its innovation rents is diminishing very rapidly (Fu et al, 2010). Initially, research and development and manufacturing moved abroad to tap talent or reduce costs but firms still focused mostly on products for wealthy nations. That is changing by empowering R&D centres in emerging markets to take greater control and develop for home turfs. An example is seen in General Electric's development of low-cost and portable ECG and ultrasound machines in India and China with eventual transfer to Western markets in what is called "reverse innovation" (Immelt, Govindarajan, & Trimble, 2009).

"Reverse innovation" picked up dramatically following the publication on "How GE is disrupting itself" by CEO of General Electric Jeffrey Immelt and colleagues Govindarajan and Trimble in 2009. They describe reverse innovation as the process of starting with the needs of low income mass consumers in developing countries to produce solutions which have global application. However Govindarajan and Ramamurti (2011) point that the focus on 'reverse' is

specifically on the flow of innovations from poor to rich countries as compared to the predominant and historical flow from rich to poor.

“Inclusive innovation” and “BOP innovation” have gradually been on the rise since the early part of the 21st century when Hart and Prahalad (2002) became known for the potential to reduce poverty and profit from the poor by treating them as self-respecting customers. Extreme affordability grew from the establishment of the "Design for Extreme Affordability" programme at Stanford University in around 2004. Although all concepts are more similar to each other than different, we focus on "frugal innovation" since we believe it more accurately captures ideas espoused by 'reverse', 'BOP', 'inclusive', 'extreme affordability', 'catalytic', and 'trickle up' or 'bottom-up'.

Frugal engineering predated frugal innovation to describe in somewhat the same way as reverse engineering, norms of engineering practise in India businesses to lower the cost of product development and manufacturing. Carlos Ghosn, CEO of Nissan and Renault, says "*Frugal engineering is achieving more with fewer resources. The cost of developing a product in the West is high since engineers there use more expensive tools. In India, they achieve a lot more with fewer resources.*" (Nair, 2008: online). Athreye and Kapur (2009: 214) contend, "*some firms from India and China have acquired a niche in "frugal engineering"—the ability to manufacture low cost versions of goods for mass markets.*" For emerging markets and developing countries, both the cost to innovate and the cost of the innovation is a primary concern. The pharmaceutical industry “over-engineered” the R&D process and even such businesses which focus on performance value or relational value should also include cost innovation as part of their strategies (Ryans, 2009, p.220). In mainstream practitioner circles ‘frugal engineering’ was highlighted by a Booz and Allen Strategy+Business publication (Vikas Sehgal, 2010) around the same time by The Economist (Woolridge, 2010).

Frugal innovation is growing in popularity as a buzz word in many practitioner and some academic circles. Don Hoefler, a journalist, popularized the term “Silicon Valley” in 1971 in a trade publication. In much the same way the term “frugal innovation” has been popularized in a special report in The Economist (Woolridge, 2010) on innovation in emerging markets. It is particularly gaining popularity in Asia’s emerging markets (EMs).

Consistent economic growth among the emerging markets serves as a complex driver for evolving innovation practises. Since the recognition of emerging markets (Van Agtmael, 1982), those economies have been growing at a significantly higher GDP growth rate than the United States and other advanced economies. The most recognizable are Brazil, Russia, India, and China (BRIC). But the World Economic Forum labels the Next 11 (N11) to gradually emerge as another set of markets to contend with. Public optimism in the economies of emerging markets fares beyond the 50% mark whereas in the OECD countries figures are far below around the 10-20% mark (Woolridge, 2010).

The Economist defines “*frugal innovation or constraint-based innovation is not just a matter of exploiting cheap labour (though cheap labour helps). It is a matter of redesigning products and processes to cut out unnecessary costs*” (Woolridge, 2010). Gupta and Wang (2009) explain frugal innovation is in deep contrast with products, services, and processes which are typically energy inefficient, raw material inefficient, and environmentally inefficient, often produced in the advanced developed economies. A notable example of frugal innovation is Aravind’s intraocular implants at \$5 compared to an import cost of \$200/unit (Prahalad, 2005). Others are the \$100 One Laptop Per Child PC, \$2000 Tata Nano car, \$1000 GE portable ECG machine, \$20 Jaipur knee, \$5K Awami (people’s) Villas, Embrace infant warmer, and in services Husk Power off-grid energy, Telenor EasyPaisa and EKO mobile phone banking.

Varadarajan (2011) believes the following drivers are helping to define the type of innovation being practiced in India: Affordability, availability, accessibility or usability, increase in

MNC RnD centres, increase in social entrepreneurs, legacy policy failures, regulatory compliance imperative, and sustainability imperative. While presented for the case of India, affordability, accessibility, availability, and sustainability are common themes relevant to most emerging and developing markets and pose common issues that frugal innovation seeks to address.

Differences in social demographics and physical infrastructure in developing countries means that consumers there are more than happy with solutions that deliver decent performance, say at 50-70% performance but at radically 10% of the cost of a full-fledged high-tech solution. By doing away with the high end product features which matter to the top 10% of population but account for 90% of the cost, one can develop products that meet the needs of 90% of the population at 10% of the cost (Woolridge, 2010). A new price-performance paradigm means that while in developed countries performance matters most, followed by features with generally little concern for price, in emerging markets price matters most followed by portability and ease of use (Immelt et al, 2009). But it isn't just cost, it is localization as well as through bottom-up strategies, or what Immelt et al call, zero-based. Unique requirements due to low purchasing power, education, scale, infrastructure, and sustainability mean that innovations have to be developed and not simply adapted.

Publications in academic, practitioner, and policy related circles seemed to have picked up following mid-2010 (figure 2). Although there is no agreed upon definition for frugal innovation, we believe “frugal innovation” has greater potential to better capture the bits and pieces espoused by complementary definitions mirrored in inclusive, reverse, extreme affordability, trickle-up, BOP, and catalytic proponents. All these concepts are growing in academic and practitioner literature as shown in figure 2. We attempt to assess the potential of specifically “frugal innovation” to emerge from fad, to fashion and possibly fit.

FAD, FASHION, OR FIT?

Is frugal innovation a fad or fashion, and can it become a fit for organizational implementation? To evaluate whether frugal innovation has future potential, we need to cover the difference between fad, fashion, and fit. A fad is a new concept or practice that bursts on the scene but then fades away after a short period of time (Hesseling, 1984; Wiele et al, 2000). Based on institutional theory, Abrahamson (1991) offers that a fad diffuses across organizations which belong to a group, i.e. organizations within a group imitate other organizations within that same group via processes of social interactions. The process of contagion is used to explain this burst in popularity where each adoption makes subsequent adoptions more likely (Abrahamson, 1991). For this article's discussion, we extend the preceding definitions based on the findings by Miller, Hartwick, and Le Breton-Miller (2004) to mean that a fad fades away almost as quickly as it became popular without leaving a mark on organizations or having been afforded evaluation to reveal evidence of its positive or negative impact.

A fad changes to a fashion if a fad is adopted widely based on some projected evidence that it seems to work (Hesseling, 1984; Wiele et al, 2000). Extending Abrahamson's (1991) assumption of a fad, fashion transpires if organizations from outside the initial group of adopting group begin to widely adopt the concept. And they do so by imitating administrative models promoted by "fashion-setting organizations" (Hirsch, 1972) whose missions involve the creation and dissemination of such models, and they do so by citing some evidence of successful adoption. These promoters reside outside the adopting organization's group and include for instance consulting firms, business mass media, and business schools (Mintzberg, 1979; Kimberly, 1981; DiMaggio & Powell, 1983; Hirsch, 1986; Meyer, 1992; Strang & Meyer, 1994) but also academic gurus and hero managers (Huczynski, 1993). Yet still Abrahamson (1996: 254) defines management fashion as "transitory collective beliefs that certain techniques are at the forefront of management progress". They are

deemed transitory because fashionable concepts or practices touch only the surface without making significant changes to the structure or practices of an organization (Wiele et al, 2000).

One reason why a fashion may only touch the surface is that the literature proposes that organizations often gain legitimacy by imitating the adoption decisions of organizations that have reputations higher than their own (DiMaggio and Powell, 1983; Fombrun and Shanley, 1990). Abrahamson and Fombrun (1994) called this a trickle-down fashion process whereby managers of higher reputation organizations adopt management fashions to distinguish their organizations from lower reputation organizations. However, lower reputation organizations follow suit and begin the adoption of the newer practices. Concomitantly, Abrahamson (1996: 272) propose that “new management fashions will tend to emerge when old management fashions have been adopted by lower reputation organizations”. Hence firms adopt management fashions without much thought on which decisions or practices they should adopt for a better "fit".

Finally, a fit occurs when the original fad turned fashion gains organization-wide commitment and becomes incorporated into the everyday working fabric of the organization and effects its overall management system (Hesseling, 1984; Wiele et al, 2000). For a fad to survive it needs to bring about organizational change which is not easy given the resource and power imbalance such change will trigger. So few fads ever progress to the stage of fit.

Given that there has not been much developments in innovation beyond the disruptive, modular, architectural innovations of the 1980s, the user and user-centric innovation of 1980s (von Hippel, 1986), distributed innovation of 1990s (Ghoshal & Nohria, 1991) and open innovation of 2000s (Chesbrough, 2000), now may be the opportune moment for newer trends in innovation to emerge. The growth trend in user, distributed and open innovation publications are shown in figure 1 and though they remain to be on the rise, it has a decade since popular innovation concepts have emerged. There seems to be a demand among managers to need new ideas on a regular basis, without knowing how the new concepts will affect performance (Wiele et al, 2000). Wiele et al

provide the analogy of the demand for gradually rounded car bodies to describe the process when management techniques become fashionable, they focus managers' preferences and further generate preferences for related management fashions. We see the dozen phrases under “What” on table 2 that capture some purposeful effort to discover new ways to innovate and create direct value for under-served populations. Fashion setters in turn recognize this managerial mood and promote related management concepts thereby triggering a management fashion trend (Abrahamson, 1996). We believe this emerging fashion trend is captured by a growing recognition of ‘frugal innovation’ in the mainstream academic and practitioner literature.

Wiele et al (2000) cite frustration, boredom and striving for novelty, and striving for status differentiation as important signs for the changing demand for fashionable concepts. This frustration and boredom can stem from a variety of technical-economic forces which in turn influence fashion demand. They cite the Japanese competitive challenge and the relative decline of U.S. and European businesses in the 1980s that generated a source of collective frustration for U.S. and European managers. This in turn fuelled a new management fashion demand (Mitroff & Mohrman, 1987) that helped with the wave for total quality management (TQM).

Fads are simple, yet vague and open to many meanings in different settings. Consequently, it becomes difficult to define them in ways which help to test out their potential. Since many definitions are possible, this helps the fad to survive for as long as each new definition is sought by managers to help them understand what they have to do in order to be seen as following the overall philosophy (Wiele et al, 2000). TQM is termed as both a fad (Wiele, et al, 2000) as well as an overall management fashion trend focused on quality enhancement. Many concepts fall within the TQM purview (Wiele et al, 2000) such as quality circles, statistical process control, failure mode and effects analysis, quality function deployment, design of experiments, benchmarking, and self-assessment. These tools, techniques, systems, and practices that compose TQM are attempts to define the meaning and to understand what needs to be done to put TQM into practice. One can

argue that TQM is the management fashion trend while some of the more transient tools that seek to define or measure it as fads.

Management literature targeted towards managers includes the Booz and Allen Strategy+Business and McKenzie Quarterly from consultants, the Economist and Business Week from journalists, and even the Academy of Management Perspectives (previously Executive) from academics. Barley, Meyer, and Gash (1988) based on a study of how corporate rhetoric spreads between the academic and popular management press, found that corporate culture rhetoric that developed in the press actually shaped the academic rhetoric, rather than vice versa (Abrahamson, 1996). In the case of "frugal innovation" specifically, the rhetoric stemmed from the popular press and particularly from a special report on innovation in emerging markets by The Economist (2010). Hirsch (1972: 643) argued that we should regard "the mass media in their gatekeeping role as a primary institutional regulator of innovation". Further Hirsch claims (649) that "the diffusion of particular fads and fashions is either blocked or facilitated at this strategic checkpoint". Subsequent to this publication we see a steep increase in the both academic and media reports on frugal innovation. Frugal innovation it seems has broken past this checkpoint. The Economist alone has published several articles on the topic following the special report (e.g. Economist, 2011b, 2011c, 2011d; Economist, 2011a, 2011e). Whether frugal innovation can help materialize the results it purports to espouse is another question. Firms may not call it as such, but many have pursued or announced strategies around frugal innovation concepts.

Following Abrahamson (1996), a lot of the literature on new popular management concepts is based on the transient character of management fashions on management fashions (e.g. Abrahamson & Fairchild, 1999; Carson, Lanier, Carson, & Guidry, 2000; Miller, Hartwick, & Le Breton-Miller, 2004). However, there is also a counter realization that management fashions may have an enduring effect (e.g. Benders & Van Veen, 2001; Clark, 2004; Scarbrough & Swan, 2001).

This enduring effect may be positive or negative. Fads can at times have a lasting destructive influence if through their pursuit, it alienates employees or triggers an abortive and expensive reorganization. Horrid examples of "downsizing" and "reengineering" show how the practices demoralized the workforce and made vital talent redundant. For this reason, Miller, Hartwick, and Le Breton-Miller (2004) deem it useful to be able to spot fads early on and to evaluate their usefulness in light of the one's needs, values, and contextual situation. We use Miller et al's (2004) criteria for fads to evaluate whether frugal innovation can be considered a contemporary fad. These eight criteria include being too simple and straightforward, promise phenomenal results, universally applicable, have step-down capability, in tune with zeitgeist, being novel and not radical, gain legitimacy thru gurus and star examples, and presented in lively, entertaining narration. From table 3 we argue that frugal innovation only conforms to three out of eight fad criteria. This supports the notion that it has broken past the fad state and into the fashion state.

Though the term's "fad" or "fashion" may seem dismissive, they are not: Concepts like Management by Objectives, or Japanese Management, TQM, and ISO 9000 can help legitimating a firm's membership and can introduce useful ideas or spin-off ideas that companies benefit from even as the fads and fashions fade away. Fads and fashions may benefit organizations if they, even if symbolically, project an image of innovativeness (Nystrom & Starbuck, 1984). And fads and fashions may involve processes that help trigger random variations from which increasingly efficient innovations can evolve (Anderson & Tushman, 1991; Hannan & Freeman, 1977). Therefore, Wiele et al caution that managers should not outright reject fads for the fads signal issues, problems, and techniques that managers care about and may lead to positive impact. Fads suggest approaches of value pertinent to specific contexts and enterprises and thus play a role in drawing attention to problems and solutions that have long remained overlooked. Wiele's et al (2000) research findings reveal that a fad in itself does not change organizations and nor is a fad

good or bad, rather it depends on how organizations implement them in light of their respective situations.

Even if frugal innovation is a fad or fashion, organizational theorists still cannot conclude that fads and fashions necessarily harm organizations (Miller et al, 2004). And even if there may be little direct economical benefits, there may exist symbolic and political efficiencies. For instance, notwithstanding the efficacy of "quality circles", they may have refocused attention on the notion of quality and worker participation (Miller et al, 2004) and "strategic planning units" may have helped focus attention on the importance of overall strategic planning (Porter, 1989). Similarly, adoption of a frugal innovation strategy may offer expressive functions which have intrinsic value and indirect benefit. A strategic choice of frugal innovation that addresses not just the high-paying clients, but also the underserved clients can make an organization appear more innovative or ethical. This may boost workforce morale and better customer participation at all market segments, and thus may help the firm raise both economical and human capital.

Frugal innovation has arisen not from the writings of academics or experts but out of practitioner responses to unique economic, social and competitive challenges faced by firms in emerging markets. There are no generally accepted guidelines or simple rules for all to follow on how frugal innovation can materialize any perceived or promised results. Frugal innovation, as it stands, seems to be complex, multifaceted, and can be interpreted and applied in a number of ways in different businesses. In much the same way as classics like globalization and outsourcing (Miller et al, 2004) have elicited research into a multitude of strategic, corporate, and functional issues, it is expected frugal innovation will have to go through an elongated, yet necessary process over time to be able to change the way a large number of companies do business.

Trends in management fashion which establish a good "fit" take longer to become widespread because of the significant human and material costs required to have a great impact on the firms that choose to adopt them. For frugal innovation to really have an impact, it will therefore

not be something on the tip of everyone's tongue, only to those that can truly commit to its implementation and value generation. As with other classics, this would mean that it will take much longer for it to go out of fashion (Miller et al, 2004).

THE TIMING OF FRUGAL INNOVATION

Why the talk on frugal innovation now? The answer may lie in changes in fashion demand triggered partly by macroeconomic fluctuations (Barley and Kunda, 1992) and partly by unsolvable contradictions originating from within organizations (Abrahamson, 1996; Blau, 1971). Managers seek to learn about management concepts and techniques that help them better cope with possible organizational performance gaps triggered by external technical and economic environmental as well as internal changes.

The internal changes may be reflected by techno-economic contradictions between employee motivation versus control, product cost versus quality, and in-sourcing versus out-sourcing which cause pendulum swings in fashion demand (Abrahamson, 1996). They may also be reflected by the existence of political forces that mark neo-Marxist accounts of contradictory, inherent, irreconcilable, and structural conflict between managers and workers. In this political conflict and control approach, management fashions may be forms of control that emerge to suppress this conflict (Abrahamson, 1996). However, Barley and Kunda's (1992) study of labor unions and strikes did not reveal coincidence with the cycles of managerial rhetorics.

External changes have been described by long-term, 50-year Kondratieff waves of economic expansion and contraction (Kondratieff, 1926; Schumpeter, 1934, 1935), which according to Barley and Kunda (1992) parallel broad changes in rational and normative managerial rhetorics and reflect managers' preferences for different types of management rhetorics. They find that rational rhetorics which stress the efficient use of structures and technologies surge when profitability seems most tightly linked to the management of capital as in during the upswings of longwaves. Normative

rhetorics which stress employee relations surge when profitability seems to depend more on the management of labor as in during the downswings of longwaves.

“Kondratiev Waves” (1925) are the notion that economic growth and decline occurs as a series of long waves each lasting around 30 to 50 years. The precise mechanisms and timing of the waves are contented, but the waves are often associated with parallel changes in both technology and economic paradigms (Dicken, 2007). Each wave is marked by growth (prosperity), decline (recession), stagnation (depression), and revival (recovery). The earliest wave is thought to have begun in early 1800s with the technological onset of the steam engine, moving into those brought about with the railways (1850s), electrical engineering (1900s), automobiles (1950s), and the current fifth wave with the advent of information technology (1990s).

Information and communications technologies have enabled global information networks. James Gleick (2011) in his book on *"The Information: a History, a Theory, a Flood"* analyses how information and related technologies have become a driver of the late 19th and early 21st century innovation economy. Thanks to the ICT revolution increased access among the poor to satellite TV and information through greater communication means that a consumer in a rural area is becoming more literate and aware of the various products that are on offer in the marketplace (Prahalad and Hart, 2002).

Though different innovations in production, distribution, and organization accompany technological change, technology alone is not a sufficient cause of economic growth (Freeman, Clark, & Soete, 1982). So researchers believe each wave brings about system-wide changes in finance, production and management (Perez, 1985). The rhetorical discourse on innovation for underserved markets reflects transformational changes in finance through mobile banking and microfinance (Klein and Mayer, 2011; Mas and Mayer, 2011) as well as in management techniques from emerging markets as those once espoused from Japan.

The rhetorical discourse in management reflected and in turn may shape the outcomes and processes in longwave eras. The rhetoric of Industrial Betterment (1870-1900) reflected the socioeconomic context of the late nineteenth century. The rhetoric of systems rationalism (1955-1980) reflected the imagery and techniques around the technical progress of computers.

The late 1800s went through great social change through revolutions in technology, mass manufacturing, corporate growth and consolidation (Hounsell, 1984). Labor relations worsened, strikes and lockouts were common and radical labor unions emerged from the immigrants' European notions of socialism. Industrial owners fearing social upheaval looked to reformers who maintained that modifying industrial conditions would alleviate the employment relations threat to society. The current "Occupy Wall-Street" and "99%" movement reflects today the same strains witnessed at the end of the 19th century.

Frederick Taylor's (1911) Scientific Management (1900-1923) perspective provided a more rational though "partial solution of the labor problem" (Taylor, 1895). Taylor actively criticized welfare capitalism's betterment programs and their notion of charity (Barley and Kunda, 1992). "No self-respecting workman," Taylor wrote, "wants to be given things, every man wants to earn things" (Taylor, 1903: 1454). Prahalad's exhortation to corporations to 'treat the poor as customers' mirrors Taylor's sentiments. Frugal innovation's affordability clamor resonates not only in bringing the price of solutions low enough to be affordable, but also increasing the earnings and savings power of consumers, and for the empowerment role a buyer enjoys over being a recipient of free hand-outs.

The organizational culture and quality discourse (1980-2000) came about with the concern of Japanese competition and the frustrating performance of Western managers. Practitioner literature sparked immense interest with cover stories in *BusinessWeek* (1980) and *Fortune* (Uttal, 1983) and through books on "Theory Z" (Ouchi, 1981), "In Search of Excellence" (Peters and Waterman, 1982), and "Corporate Cultures" (Deal and Kennedy, 1982). Efforts to revitalize Western industry in the face of Japanese competition saw the "Total Quality Movement" and the

movement for "World Class Manufacturing." Barley, Meyer, and Gash (1988) discovered that by the mid-1980s the practitioner-oriented view had become dominant, even in academic circles and not vice versa.

The frugal innovation discourse has too emanated from the practitioner community beginning around 2008 espousing innovation by Asian firms (Adolph & Pettit, 2008; Brown & Hagel, 2005; Brown & Hegel, 2005; Chaudhuri, Giffi, Kandaswami, & Singh, 2009; Ernst&Young, 2011; Jaruzelski & Dehoff, 2008) and by the emerging market giants mainly China and India (Economist, 2010). With practitioner literature first outlining the frugal innovation space, and now subsequently being followed by research and empirical concerns by the academic community, here too we may witness the dominant view to be defined by the practitioner community. If that turns out to be true, the relevance of frugal innovation according the (Wiele et al, 2000; Miller et al, 2004; Barley, Meyer, and Gash, 1988) may become less fad-like and more in fashion as managerial discourse. However, to make it a good "fit" for organizations to effectively imitate or adopt notions of frugal innovation strategies, academic research is needed to help define its full potential as well as limitations.

THE BEGINNING OF THE NEXT KONDRATIEV WAVE OR BUSINESS CYCLE?

Freeman (1987) argues that each of these Kondratiev waves or 'techno-economic paradigms' requires a fundamental restructuring of the socio-institutional framework. The innovative capacity of an innovation system depends on the cooperation between varied activities and actors such as education for expertise and training, intellectual property for financial returns, and entrepreneurs for commercialization (Freeman et al., 1982). Perez (1985) cites the social, political and institutional transformations as responsible for growth and shape of the next longwave.

Take for instance the examples of mobile phone banking and microfinance. Richard Sylla's study of financial systems (2002) offers that innovative "financial revolutions" predated the growth and success of leading economies of modern times. Sylla posits that viable financial systems have a

causal role in economic modernization defends by highlighting crucial institutional components of financial systems of the American, Dutch, and British societies. The Dutch financial revolution took place in the first decades of the seventeenth century, and British financial revolution occurred in the late seventeenth century, and the US financial revolution occurred in less than a decade after the ratification of the constitution (Sylla, 2002). All these predated the rise of their respective economic golden ages.

Colin Mayer (2011: video) says *"a revolution is taking place in finance, it is a revolution that is not happening in the streets of New York or London or New York, but in the slums of Kenya (through) what is termed mobile banking"*. According to the International Monetary Fund, in October 2011 M-Pesa conducted more transactions domestically within Kenya than Western Union does globally. The project began in 2003 as an initiative of the UK development agency, DFID, with Vodaphone. On comparing microfinance with mobile banking, Mayer believes mobile banking will be even more exciting than microfinance because instead of placing emphasis on borrowing and people getting into debt, mobile banking can help people accumulate savings.

Most diffusion spreads as per the classic "S-shaped" diffusion curve which is characteristic of contagion processes (Simmons et al., 2006). Technologies, organizational forms and institutions and regulations such as privatization, financial openness, and democracy have all followed this "S-shaped" ferment, growth, and decline phases. When growth slackens due to saturated demand or shrinking profits given intense competition, investment falls, firms restructure and unemployment rises. Following the lowest point of the wave, a new sequence begins on the basis of these new technologies and on new investment opportunities (Dicken, 2007).

We think some of these new technologies may be emerging in the form of frugal innovations. Indeed some such as leaner and simpler banking models like *Simple.com* and *Mint.com* and collaborative consumption models like *Zipcar.com* are based on innovations triggered at times of recession. The growing trend of collaborative consumption where car (RelayRides), house

(Airbnb), and baby clothes sharing (Plumgear) are brokered by web start-ups in San Francisco are in many forms and practices frugal innovation. These start-ups leverage existing assets owned by the public to reduce their own start-up capital, enable providers to earn income from idle assets, and allow consumers to avail services at lower cost as compared with traditional business models of car and hotels rentals. It is difficult to predict a new longwave, but the signs embodying development of several new socio-techno-economic systems indicates a new Kondratieff wave may have already begun.

THE EMERGENCE OF THE MARKET FOR FRUGAL INNOVATION

A market is a set of institutional arrangements that allows different actors to come together to form means for value creation. In a report on "Innovating for the next three billion" by Ernst and Young (2011), companies will have to create entirely new products and services to fulfil the unmet needs of the three billion new customers. We extend that to mean that those customers are not going to simply be 'out there' to be tapped, but rather that new markets have to be generated for the three billion to be inclusively served.

The evolution of markets is broadly explained through either a technology push or socio-political pull strategy. The former approach stems from Geroski's (2003) studies in industrial economics and technology innovation arguing that technological innovation drives market formation. This process is supply-side driven, path-dependent and the demand-side only matters in choosing the dominant design. The latter approach stems from Fligstein's (1996; 2002) studies in history of technology and economic sociology arguing that instead market formation is a kind of social movement process of political claims-making, cultural traditions, and boundary definition by institutional actors including state agencies (Santos and Eisenhardt, 2009; Weber, Davis, Lounsbury, 2009; Hargadon and Douglas, 2001). This political-cultural process is charted by what Hughes (2004) calls 'system builders' who help to assemble distributed elements of the political-

cultural-technological system into a coherent system of innovation that spreads through its own momentum.

So the innovation literature argues broadly either the technology push strategy for meeting inchoate demand (Geroski, 2003) or the pull strategy through user centric innovation (Von Hippel, 1986). We believe neither fully explains the rise of frugal innovation. Instead, the social movement lens in which actors shape discourse and agendas more reliably, we believe, demonstrates the emergence of the market for frugal innovation. The market for frugal innovations is at a formative fluidic stage with growth yet to be seen. We draw from these two approaches into conversation to map out a process-rich account of the rise of the market for frugal innovations and their spread across the wider global space. A history of the processes leading to an event's occurrence are as interesting as understanding the culminating event in history itself. We have shown above the range of complementary and contending 'whats' that describe the frugal innovation space. We find a cluster of activities in the 'how' space where some are focusing on reducing costs, some are redesigning value chains, and others focusing on building local capacity (table 2). But the 'whys' that hold these together are far fewer and help to coalesce the varied actors towards the mission oriented goals of fulfilling needs of the underserved whether for morality, profit, or social impact.

We extend Fligstein's view of "markets as politics" to argue that the space for frugal innovations is being charted as a socio-political contest by various actors such as political leaders, business elites, academic scholars, and even religious clergy to confront extraordinarily constrained economic and political environments which are characterized by burgeoning unmet yet basic demands. In studies of the emergence of financial growth markets in Europe, Posner (2004 & 2009; Weber and Posner, 2000) argues that rather than because of the much touted efficiency of these markets, it is instead politics that more importantly shaped the processes of experimentation and drove the diffusion of these alternative markets across Europe.

Fligstein's (1996) approach of "markets as politics" implies that states play an important role in how the market evolves.

States are expected to provide equitable conditions, under which firms function to organize, cooperate, compete, and exchange. For instance, property rights, rules of exchange, and governance and remediation structures help to establish rules for economic actors (Fligstein, 1996). But laws formulated for this purpose are rarely neutral as they favour certain groups of firms (Fligstein, 1996). In turn, political discourse and contests define the laws, their applicability, and extent of penalty or exclusion. Markets in the formative stage are akin to a state of crisis and that resembles a social movement.

King and Pearce (2010) demonstrate how social movement theory serves as an explanation for the emergence of market heterogeneity (Schneiberg & Lounsbury, 2008). Rucht (1999) defines a social movement as "*an action system comprised of mobilized networks of individuals, groups and organizations which, based on a shared collective identity, attempt to achieve or prevent social change, predominantly by means of collective protest*" (p.207). Social movement literature has broadened from movements triggered by the poor and frustrated masses to resource and capabilities empowered mobilization by the middle classes and beyond to the new social movements fostered by elites who shape discourse and agendas.

Social movement research shows empirically that contentiousness is a source of market dynamism, for instance by encouraging innovation and by changing what is deemed acceptable practice (Rao, 2009). The rise of the market for organic and grass-fed beef and dairy (Weber, Heinze, & DeSoucey, 2008) is attributed to social movements helping to create consumer audiences for these once ignored products (King and Pearce, 2010). We therefore are borrowing the argument that social movements form markets, that a coalition of academics, idealistic students, global policy people, business elites, funders, foundations, and ventures are collaborating, contending and

negotiating to serve unmet needs for the world's underserved. The major players are shown in figure 3.

Insert Figure 3 here

ACTORS DEFINING MARKET FOR FRUGAL INNOVATION

To frame this work in terms of the 'market' for frugal innovation, we want to assemble the institutional arrangements, actors, and processes that are building this market. In the same sense as a market for corporate governance' where it revolves about the institutional components of accountability, we propose that the 'market' for frugal innovation involves the alignment of key institutional players around some converging set of concepts and the spill-over of this alignment into policies and local practices.

A host of actors comprising development economists, policy makers, educationists, idealistic students, foundations with a social mission, investors seeking both profit and social impact, and bottom of the pyramid consumers potentially voicing frustration with governments' inability to meet their needs are together making up a social movement for addressing poverty and unmet needs.

Development economists such as Sachs, Easterly, Collier, and Pollock have debated the public and market approaches to poverty reduction. World leaders have committed to the UN's Millennium Development Goals to eradicate poverty by 2020. Educational programmes such as Santa Clara University's Global Social Business Incubator and the Frugal Innovation Lab and Stanford's Design School Extreme Affordability classes are training the next cohorts on developing products and solutions for the underserved. The Bill and Melinda Gates and the Skoll foundation are but a few of the non-profit foundations seeking to catalyse this market formation. Venture capital investment funds for social impact have only recently formed in the last decade to fill the gap in resources for social causes sought to be solved by entrepreneurs. Social impact investment funds include the Acumen Fund, Omidyar network, and the Mulago foundation to name a few all of

which have materialized only within the last decade. Global companies such as General Electric and TATA have embraced frugal innovation missions and strategies for differentiation and / or competitive advantage. They are influencing 'less reputed' firms to follow suit in a fad-fashion trend. Entrepreneurs and non-profits work in extreme environments to fulfil needs of regions often neglected by state agencies and are approaching new forms of venture funds who seek to invest with social impact. Consulting firms positioned in emerging markets propagate emerging market social, demographic, and business trends either as opportunity or competitive threat.

In this multidimensional space some maintain cost reduction is the goal whether done through local knowledge, skills, and materials or through optimizing international supply chains while others hold that local capacity building is the ultimate mission. The debates between optimization and local capacity generation mean some compromise either today or tomorrow. Efficiency today may mean losing capacity tomorrow and capacity and independence today may mean losing efficiency tomorrow. For instance, outsourcing is good today for efficiency, but over 10 years, you may lose the capacity to do it yourself.

As an example of the contention in this field, consider the antecedent to frugal innovation, 'appropriate technology'. Appropriate Technology is based on Schumacher's "Small Is Beautiful" book (1974). Schumacher criticised the neoclassical emphasis on blueprints, mass production and transfer of technology to developing countries, arguing that such an approach did not make economies sustainable. In contrast, Schumacher stressed the need for technologies to be appropriate for local conditions, to be adapted to current economic and resource conditions and to avoid human displacement or environmental degradation (Sonne, 2010). But until very recently, this notion has been unpacked and repacked to form new techniques on the 'how' to do it. For instance, Kickstart has reconfigured its delivery model away from localized manufacturing to supply chain efficiency, given what they deem to be the failure of appropriate technology to scale up in meeting unmet local needs.

Barley and Kunda (1992: 381) described the shift from systems rationalism to normative organizational culture as "commitment was to quality as calculation was to efficiency". We offer that frugal innovation is marked today by a "concern to social inclusion and equality". In some states, norms of inclusiveness create a frugal innovation market for solutions championing inclusive progress. For instance in India, the Indian Planning Commission has included in its draft twelfth five-year plan to include frugal innovation as a technology and innovation strategy for more inclusive growth and development:

"Innovation can fulfill needs, which are not met by conventional means and this is critical in view of the numerous challenges the country is facing in delivering services to the people, especially at the bottom of the pyramid. In this context India needs more "frugal, distributed, affordable innovation" that produces more "frugal cost" products and services that are affordable by people at low levels of incomes without compromising the safety, efficiency, and utility of the products. The country also needs innovation processes that are "frugal" in terms of resources required and these products and services must also have a "frugal" impact on the earth's resources and must be designed to be environmentally sustainable. Innovations in India hence need to cast a wider net to benefit more and more people who are currently marginalized by the system." (Planning Commission, 2011: para 2)

Several movements together have brought awareness to the development of inclusive markets for the poor in developing countries. The environmental movement saw its environmentally oriented activities popularized by the first Earth Day on April 22, 1970 (Lounsbury, Ventresca, and Hirsch, 2003). In many ways similar to how current inclusive social causes are championed first by social enterprises, in the 1970s, recycling activities were opportunities to provide less well-off individuals with jobs (Weinberg, Pellow, & Schnaiberg, 2000) and to build self-sustaining communities (Lounsbury et al, 2003). Social movements against globalization and the WTO sought to limit the power of global multinationals. The most recent (and current) Occupy Wall Street and the 99% movement are calling for more inclusive financial and democratic reforms.

Poverty reduction has been globally supported by the UN's Millennium Development Goals. Governments, aid agencies and NGOs have rallied around the MDGs to help the world attain equitability in the five areas of focus; health, education, environment, equality and economic development (Sachs, 2006). Concomitantly, the private sector has rallied around the market mechanism to solve these issues (Easterly, 2006; Novogratz, 2010; Polak, 2008; Prahalad, 2005). But poverty reduction is not just simply pushed from policy makers, rather is also being pulled by those who need it most.

Information availability, whether in the form of news, marketing or entertainment means consumers regardless of market segmentation are becoming aware of products and services that would otherwise enhance their lives. They are therefore demanding solutions afforded by the rich (Prahalad, 2005) and the BOP market is emerging as a large market for a number of goods or services—whether a FMCG consumer good or a financial or telecom service. Concomitantly, providers are being connected to the needs and concerns of such potential users who have historically been excluded from the practicalities of markets, whether as suppliers, producers or users. They can thus use these same technologies in their business models and strategies to shape and deliver solutions. Examples of these include mobile banking, e-health, and e-market boards.

The BOP literature for poverty reduction vacillates between large corporation led and entrepreneur led approaches. While Prahalad encouraged the former, others have debated for entrepreneurs to step up. The BOP strategies are not solely focussed on poverty but also sustainability, again two popular demands of social movements. Regardless of the critiques of Prahalad's BOP vision, Landrum (2007) believes that Prahalad's contributions include first, raising awareness of the BOP, the challenges they face, and global poverty and second, challenging corporations to increase innovation and creativity, with a special emphasis on strategies for entering emerging economies. Alternatively, where there may be barriers to the functioning of markets to contribute to the more efficient use of environmental and natural resources, entrepreneurial action

can help overcome towards the development of a more ecological sustainable economy (Dean & McMullen, 2007).

Given that a market is a set of institutional arrangements that allows different actors to come together to form means for value creation we see that a range of firms, states, academics, funders, entrepreneurs, and social activists are negotiating the terrain for the market for frugal innovation.

IMPLICATIONS AND FUTURE RESEARCH

The Bill and Melinda Gates Foundation, Skoll Foundation, Acumen Fund, Ebay-Omidyar Foundation, D-Rev, TATA, GE, Frugal Innovation Lab, Grameen Creative Lab, Global Social Business Incubator and the Design for Extreme Affordability are but a few of the foundations, firms, and university based initiatives involved in shaping the frugal innovation market. The fact that so many players are collaborating and competing to define the rhetoric around “frugal innovation” market suggests that this management fashion trend is growing quickly and poses an opportunity to understand its evolution. Future research can benefit from studies of such entrepreneurs, firms, and also the roles that business and engineering associations and professional schools are playing in shaping these efforts.

In contrast to large expenditures in public and private R&D as necessary precursors to innovation, the concept of frugal innovation herein enables recognition of a new type of innovation which debunks such heavy investment claims. Research may be able to show that too much R&D investment versus very little results in different capabilities to innovate for different contexts and markets. In contrast to diffusion of innovation theory (Rogers, 1962; 2003) which deems wealthy innovators and early adopters as the first likely set of users, frugal innovation users fall in the relatively poor early majority and late majority segments of users. These contrasts offer new avenues for further research on backtracking on the lead or early adopter users.

The trend of new entrants challenging established industry incumbents by offering simpler alternatives that are just good enough for underserved niches resounds favorably with Christensen's

disruptive innovation model (Bower and Christensen, 1995). Although still in its infancy, but could frugal innovations like the Tata Nano, mobile banking, and \$800 heart surgeries be considered disruptive innovations?

Further, as the West looks to the East for innovation, ideas, and entrepreneurship (Capelli et al, 2010; Ming-Jer and Miller, 2010), frugal innovation could offer fresh ideas and new perspectives in cost minimization and innovation maximization to a world slowed down by the recent recession, i.e. doing more with less. In some ways, the activity of frugal innovation is already catching on in the US in its own way. The current trend of collaborative consumption where car (RelayRides), house (Airbnb), and baby clothes sharing (Plumgear) are brokered by web start-ups in San Francisco who use existing publicly owned assets are in many forms and practices frugal innovation.

Finally, given the implications of emerging market frugal innovation to theory and practice, we conclude that, it has the potential to change the theoretical and economic landscape in a way that permanently alters corporate and global competitive dynamics (Monk, 2009). For instance, towards the end of last century 'lean' processes based on eliminating waste (e.g. lean engineering and just-in-time manufacturing) diffused from Japan's Toyota Production System (TPS) and into mainstream business. This century is facing emerging trends in frugal, reverse, extreme and BOP innovation, call it what you like, that uses a mix of technology, social, and institutional (Haque, 2010) innovations to create new markets, both in the East and West. This may lead to "Frugal" as the next integrated management philosophy to diffuse from the East to the West. While lean is focused on "preserving value with less work" or "eliminating waste", frugal may be centered on "creating more value with less".

The big question will be whether its diffusion will serve as a competitive advantage for adopters? Perhaps not, because regardless of the plethora of TPS guides and training consultants available, many will agree that still no one does TPS better than Toyota.

CONCLUSION

The parallel growth in emerging markets and the recessionary struggles faced by the developed markets have questioned many of our prevailing 'doing business' assumptions and led to a wave of 'rethinking business'. The growth in communication technologies, particularly mobile, internet and the accompanying layer of social media have together helped to inform users of products and solutions available 'out there' but also to help connect suppliers with consumers. Social movements have raised awareness for societal issues on climate change, economic inequality, and poverty reduction. We outlined how different stakeholders, in response to these challenges, are negotiating through cooperation and confrontation not only the 'fad-like and fashionable' rhetoric around 'frugal innovations' but in the process are building a social movement that is helping to map out the market for frugal innovation. Emerging strategies, methods, and processes that deliver to this market will have to cope with resource constraints and at the same time either maintain or improve societal, ecological and economic sustainability.

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TABLE 1: Innovations Projected as ‘Frugal’, ‘Inclusive’, ‘Reverse’ etc.

YEAR	PRODUCT/SERVICE	DESCRIPTION
1975	Microfinance	Microloans Grameen
2002	AuroraLab Lenses	\$5 intraocular lense by Aravind Eye Hospital
2004*	One Laptop Per Child	\$100 laptop for children worldwide
2004	Dacia Logan	\$5K car for Europe
2007	Mobile Phone Banking	M-Pesa in Kenya; Telenor Easy Paisa in Pakistan
2008*	Cardiac Surgery	\$800 cardiac surgery by Dr Devi Shetty’s Narayana Hosp.
2008*	Tata Nano	\$2000 Car for India’s masses
2008	Husk Power Systems	Off Grid electricity from Rice Husk
2009	Remotion Jaipur Knee	\$30 prosthetic knee
2009*	GE ECG	\$1000 handheld electrocardiogram device
2009*	GE Ultrasound	\$15K PC based ultrasound device
2009*	Zhongxing X-Ray Machine	\$20K (1/20 th price) of most x-ray machines
2009*	Tata Swach	½ dollar per month filtered water device
2010*	Gudrej Fridge	\$60 portable refrigerator for rural areas
2011*	Aakash Tablet	\$35 tablet for India’s school kids

Note: * Year indicates public disclosure in media and not necessarily year of conception or successful implementation; some are challenges yet to be met

TABLE 2: Complementary approaches to frugal innovation

WHAT	Google Scholar Hits	HOW	WHY
1. Reverse Engineering	83,127	1. Cost Efficiency	1. Profiting from new markets
2. Appropriate technology	60,860	2. Minimally Viable Product	2. Investing with Social impact
3. Lean Engineering *	18,602	3. Optimizing supply chains	3. Serving with an ethical mission (e.g. “White man’s burden”)
4. Inclusive Innovation	206	4. Rebuilding value chains	
5. Reverse Innovation	178	5. User Centric Design	
6. Extreme Affordability	73	6. Local capacity building	
7. BOP Innovation **	69	7. Creative improvisation	
8. Frugal Innovation	66	8. Business model innovation	
9. Frugal engineering	60		
10. Trickle up/Bottom up Innov	17		
11. Below the Radar innovation	17		
12. Catalytic innovation	16		
13. Innovation under constraints	9		
14. Market for underserved***	3		

Note: Search based on year-on-year end hits since 1990 and not directly from 1990-current

* Search terms: “lean engineering” OR “lean product development” OR “lean manufacturing”

** Search terms: “bottom of the pyramid innovation” OR “bop innovation”

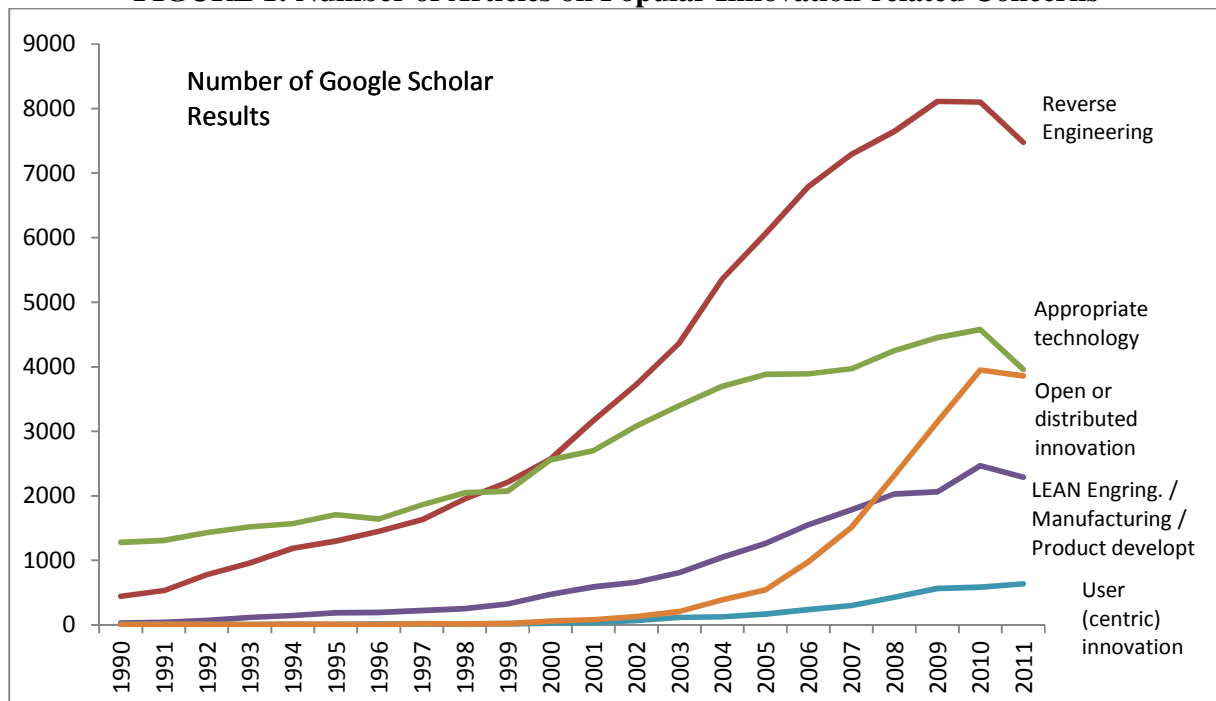
*** Search terms: “market for underserved” OR “Innovation for underserved”

TABLE 3: Assessing the Fad tendency of “Frugal Innovation”

Fad criteria	Description	Tendency
<i>Simple, Straightforward</i>	A fad’s ideas are easy to communicate, comprehend, and reduce to a small number of factors, dimensions, or characteristics. Clear-cut distinctions, perfect contrasts, and ideal types are proposed. Simple solutions are suggested.	NO
<i>Promising results</i>	Fad auteurs are confidently didactic. There is no false humility or hedging. Fads promise results such as greater control and efficiency, more motivated and productive workers, more satisfied customers, or some other valued result.	NO
<i>Universal</i>	Fads propose solutions for everyone. Imparted truths are said to apply to almost all organizations, functions, tasks, individuals, or cultures. Fads claim enormous generality and universal relevance.	NO
<i>Step-down Capability</i>	Fads have the capacity to be implemented in ritualistic and superficial ways. Recommendations can be implemented quickly and easily, often without having much effect on organizational practices. Recommendations involving large expenditures of resources or substantial redistributions of power can be avoided.	NO
<i>In tune with zeitgeist</i>	Fads resonate with the major trends or business problems of the day. They respond to challenges that are broadly felt and openly discussed. These might result from deficiencies in current administrative practices, technology changes, or shifts in economic or social conditions. Solutions are in tune with prevailing values.	YES
<i>Novel, not radical</i>	Fads are novel, not radical. They question existing assumptions, criticize widespread practices, and point to fresh new ways of doing things. However, this novelty is not so much a new discovery as a rediscovery and repackaging of older ideas, values, and approaches.	NO
<i>Legitimacy via gurus and star examples</i>	Fads are supported by tales of excellent companies and the status and prestige of gurus, not by solid empirical evidence. Stories of corporate heroes and organizational successes provide role models and suggest prestigious adherents, lending an aura of legitimacy to the ideas being espoused.	YES
<i>Lively, entertaining</i>	Fads are almost always presented in a way that can be described as concrete, articulate, bold, memorable and upbeat. They are filled with labels and buzzwords, lists and acronyms. Interesting anecdotes and corporate war stories abound. Descriptions are vivid and extreme, making fads fun to read about and listen to.	YES

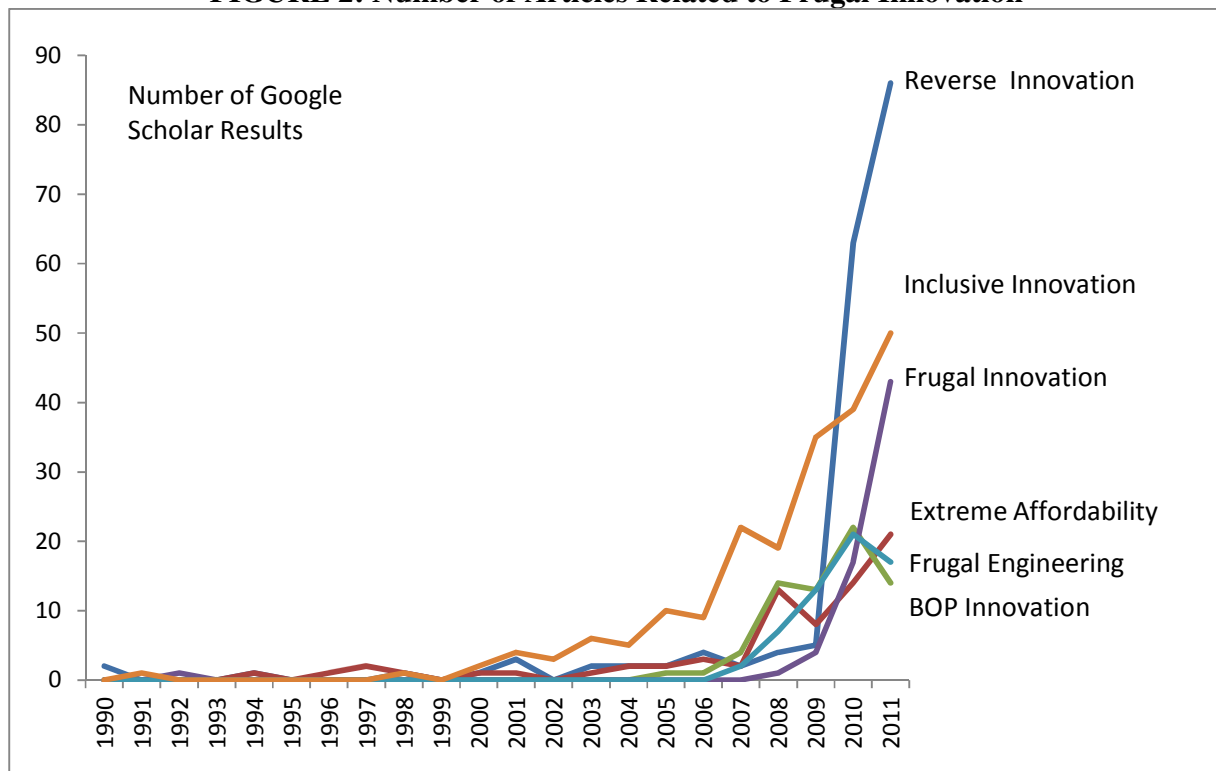
Evaluation criteria from Miller, Hartwick, and Le Breton-Miller, 2004: 11

FIGURE 1: Number of Articles on Popular Innovation-related Concerns



Note: The tapering off at 2011 is due to incomplete data for 2011.

FIGURE 2: Number of Articles Related to Frugal Innovation



Note: The rise in number of publications is despite that 2011 data is incomplete.

FIGURE 3: Major Players in The Market for Frugal Innovations

